Solutions of home work

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☑ Answer Key – Question 1

Interactive START Button

```
import pygame
pygame.init()
win = pygame.display.set_mode((720, 500))
pygame.display.set_caption("Button Demo")
WHITE = (255,255,255)
LIGHT = (170,170,170)
DARK = (100,100,100)
font = pygame.font.SysFont("Arial", 35)
text = font.render("START", True, WHITE)
while True:
    win.fill((0,0,0))

    mouse = pygame.mouse.get_pos()

    # Button coordinates
    bx = 290
    by = 220
    bw = 140
    bh = 50

    # Hover effect
    if bx <= mouse[0] <= bx+bw and by <= mouse[1] <= by+bh:
        pygame.draw.rect(win, LIGHT, (bx,by,bw,bh))
    else:
        pygame.draw.rect(win, DARK, (bx,by,bw,bh))

    win.blit(text, (bx+25, by+5))

    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            pygame.quit()

        if event.type == pygame.MOUSEBUTTONDOWN:
            if bx <= mouse[0] <= bx+bw and by <= mouse[1] <= by+bh:
                print("Game Started")
```

```
                pygame.display.set_caption("Game Running")

    pygame.display.update()
```

☑ **Answer Key – Question 2**

Keyboard Controlled Player Box

```
import pygame
pygame.init()
win = pygame.display.set_mode((720, 500))
pygame.display.set_caption("Player Movement")
x = 200
y = 200
size = 20
vel = 5
font = pygame.font.SysFont("Arial", 25)
while True:
    pygame.time.delay(20)
    win.fill((0,0,0))

    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            pygame.quit()

    keys = pygame.key.get_pressed()

    if keys[pygame.K_LEFT] and x > 0:
        x -= vel
    if keys[pygame.K_RIGHT] and x < 720-size:
        x += vel
    if keys[pygame.K_UP] and y > 0:
        y -= vel
    if keys[pygame.K_DOWN] and y < 500-size:
        y += vel

    pygame.draw.rect(win, (0,255,0), (x,y,size,size))

    # Display coordinates
    coord = font.render(f"X={x}, Y={y}", True, (255,255,255))
    win.blit(coord, (10,10))

    pygame.display.update()
```

☑ **Answer Key – Question 3**

Button Controlled Movement

```
import pygame
pygame.init()
win = pygame.display.set_mode((720, 500))
pygame.display.set_caption("Button Movement")
x = 350
y = 200
size = 30
speed = 10
LIGHT = (180,180,180)
DARK = (100,100,100)
WHITE = (255,255,255)
font = pygame.font.SysFont("Arial", 25)
def draw_button(text, x, y):
    mouse = pygame.mouse.get_pos()
    rect = pygame.Rect(x,y,80,40)

    if rect.collidepoint(mouse):
        pygame.draw.rect(win, LIGHT, rect)
    else:
        pygame.draw.rect(win, DARK, rect)

    label = font.render(text, True, WHITE)
    win.blit(label, (x+15, y+8))
    return rect
while True:
    win.fill((0,0,0))

    # Draw moving box
    pygame.draw.rect(win, (255,0,0), (x,y,size,size))

    up = draw_button("UP", 320, 420)
    down = draw_button("DOWN", 320, 460)
    left = draw_button("LEFT", 230, 460)
    right = draw_button("RIGHT", 410, 460)

    mouse = pygame.mouse.get_pos()

    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            pygame.quit()
```

```
        if event.type == pygame.MOUSEBUTTONDOWN:
            if up.collidepoint(mouse) and y > 0:
                y -= speed
            if down.collidepoint(mouse) and y < 500-size:
                y += speed
            if left.collidepoint(mouse) and x > 0:
                x -= speed
            if right.collidepoint(mouse) and x < 720-size:
                x += speed

    pygame.display.update()
```